



CHAPTER 4

Differentiation: The Amplitwist Concept

4.1 Introduction

Having studied functions of complex numbers, we now turn to the *calculus* of such functions.

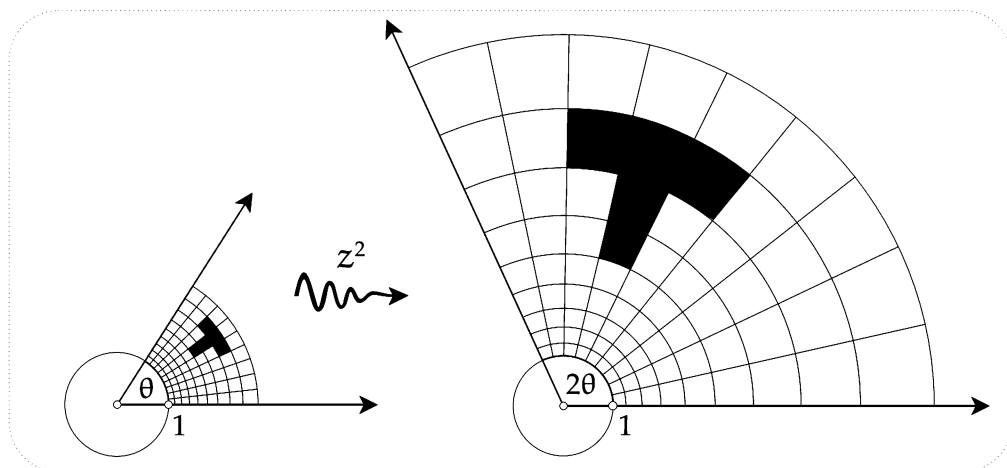
To know the graph of an ordinary real function is to know the function completely, and so to understand curves is to understand real functions. The key insight of differential calculus is that if we take a common or garden curve, place it under a microscope and examine it using lenses of greater and greater magnifying power, each little piece looks like a *straight line*. When produced, these infinitesimal pieces of straight line are the tangents to the curve, and their directions describe the local behaviour of the curve. Thinking of the curve as the graph of $f(x)$, these directions are in turn described by the derivative, $f'(x)$.

Despite the fact that we cannot draw the graph of a complex function, in this chapter we shall see how it is still possible to describe the local behaviour of a complex mapping by means of a complex analogue of the ordinary derivative—the “amplitwist”.

4.2 A Puzzling Phenomenon

Throughout Chapter 2 we witnessed a very strange phenomenon. Whenever we generalized a familiar real function to a corresponding complex function, *the mapping sent infinitesimal squares to infinitesimal squares*. At present this is a purely empirical observation based on using a computer to draw pictures of the mappings. In this chapter we begin to explore the theoretical underpinnings of the phenomenon.

Let’s go back and take a closer look at a simple mapping like $z \mapsto w = z^2$. As we already know, this maps the origin-centred circle $|z| = r$ into the circle $|w| = r^2$, and it maps the ray $\arg(z) = \theta$ into the ray $\arg(w) = 2\theta$. An obvious consequence of this



[4.1] The tip of the iceberg: small squares map to small squares—but why? Clearly the mapping $z \mapsto z^2$ preserves the orthogonality of origin-centred rays and circles. So the small, ultimately vanishing “squares” on the left must be mapped to small, ultimately vanishing rectangles on the right. Very surprisingly, though, squares are ultimately mapped to squares! As we shall see, this is a consequence of the fact that the mapping is conformal.

is that the right angle of intersection between such circles and rays in the z -plane is preserved by the mapping, which is to say that their images in the w -plane also meet at right angles. As illustrated in [4.1], a grid of infinitesimal squares formed from such circles and rays must therefore be mapped to an image grid composed of infinitesimal rectangles. However, this does not explain why these image rectangles must again be *squares*.

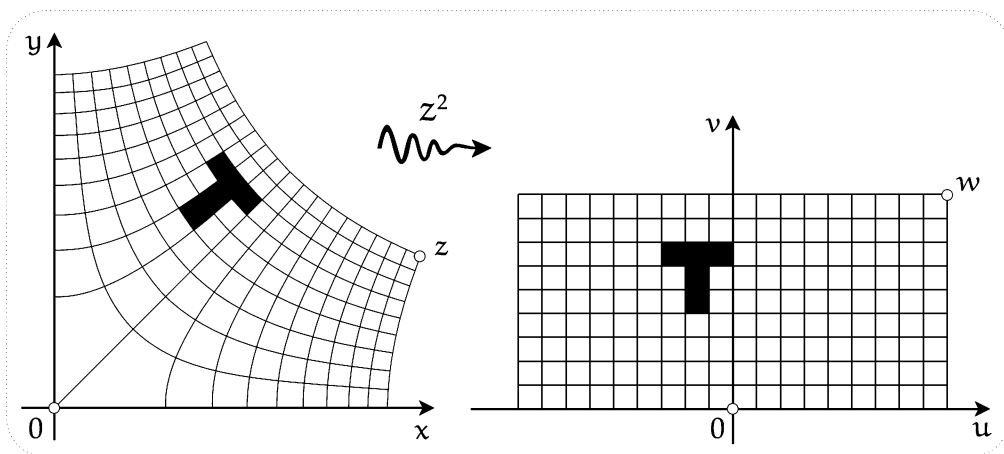
As we will explain shortly, the fact that infinitesimal squares are preserved is just one consequence of the fact that $z \mapsto w = z^2$ is *conformal* everywhere except at the two critical points $z = 0$ and $z = \infty$, where angles are doubled. In particular, any pair of orthogonal curves is mapped to another pair of orthogonal curves. In order to give another example of this, we first dismember our mapping into its real and imaginary parts. Writing $z = x + iy$ and $w = u + iv$, we obtain

$$u + iv = w = z^2 = (x + iy)^2 = (x^2 - y^2) + i2xy.$$

Thus the new coordinates are given in terms of the old ones by

$$\begin{aligned} u &= x^2 - y^2, \\ v &= 2xy. \end{aligned} \tag{4.1}$$

We now forget (temporarily!) that we are in $\mathbb{C}$, and think of (4.1) as simply representing a mapping of $\mathbb{R}^2$ to $\mathbb{R}^2$. If we let our point (x, y) slide along any of the rectangular hyperbolas with equation $2xy = \text{const.}$, then we see from (4.1) that its



[4.2] Again, small squares map to small squares—but why? The pre-images of the horizontal and vertical lines on the right are the two families of hyperbolas on the left, which a computation confirms are orthogonal to each other, as illustrated. Thus the squares on the right must have pre-images that are rectangles on the left. But, once again, surprisingly, the pre-images of the squares are actually squares!

image (u, v) will move on a horizontal line $v = \text{const.}$ Likewise, the preimages of the vertical lines $u = \text{const.}$ will be another family of rectangular hyperbolas with equations $(x^2 - y^2) = \text{const.}$ Since their images are orthogonal, the claimed conformality of $z \mapsto z^2$ implies that these two kinds of hyperbolas should themselves be orthogonal.

Figure [4.2] makes it clear that they are indeed orthogonal. We may verify this mathematically by recalling that two curves are orthogonal at a point of intersection if the product of their slopes at that point is equal to -1 . Implicitly differentiating the equations of the hyperbolas, we find that

$$\begin{aligned} x^2 - y^2 = \text{const.} &\Rightarrow x - yy' = 0 \Rightarrow y' = +(x/y), \\ 2xy = \text{const.} &\Rightarrow y + xy' = 0 \Rightarrow y' = -(y/x). \end{aligned}$$

Thus the product of the slopes of the two kinds of hyperbola at a point of intersection is -1 , as was to be shown.

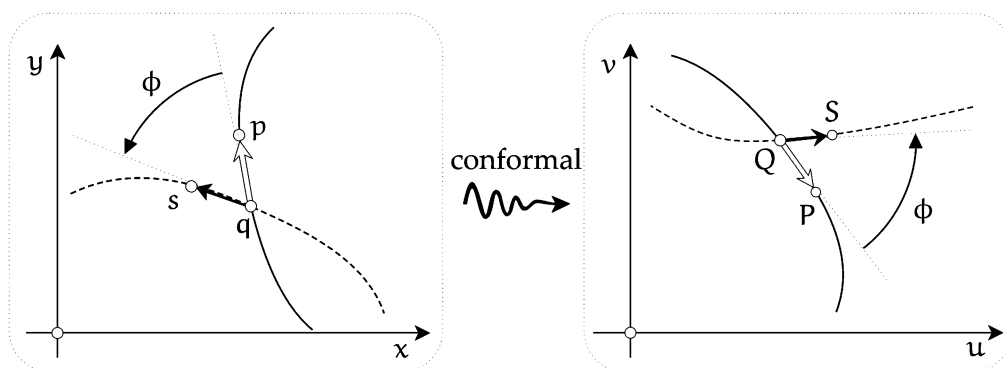
Clearly we could carry on in this way, analysing the effect of the mapping on one pair of curves after another, but what is really needed is a general argument showing that if two curves meet at some arbitrary angle ϕ , then their images under (4.1) will *also* meet at angle ϕ . To obtain such an argument, we shall continue to pretend that we are living in the less rich structure of $\mathbb{R}^2$ (rather than our own home $\mathbb{C}$) and investigate the local properties of a general mapping of the plane to itself.

4.3 Local Description of Mappings in the Plane

4.3.1 Introduction

Referring to [4.3], it's clear that to find out whether any given mapping is conformal or not will require only a local investigation of what is happening *very near* to the intersection point q . To make this clearer still, recognize that if we wish to measure ϕ , or indeed even define it, we need to draw the tangents [dotted] to both curves and then measure the angle between them. We could draw a very good approximation to one of these tangents simply by joining q to any nearby point p on the curve. Of course the nearer p is to q , the better will the chord qp approximate the actual tangent. Since we are only concerned here with directions and angles (rather than positions) we may dispense with the tangent itself, and instead use the infinitesimal vector $\overrightarrow{qp}$ that points along it. Likewise, after we have performed the mapping, we are not interested in the positions of the image points Q and P themselves; rather, we want the infinitesimal connecting vector $\overrightarrow{QP}$ that describes the direction of the new tangent at Q . We will call this infinitesimal vector $\overrightarrow{QP}$ the *image* of the vector $\overrightarrow{qp}$. However natural this terminology may seem, note that this really is a new sense of the word “image”.

Let us now summarize our strategy. Given formulae such as (4.1), which describe the mapping of the points to their image points, we wish to discover the induced mapping of infinitesimal vectors emanating from a point q to their image vectors emanating from the image point Q . In principle, we could then apply the latter mapping to $\overrightarrow{qp}$ and to $\overrightarrow{qs}$, yielding their images $\overrightarrow{QP}$ and $\overrightarrow{QS}$, and hence the angle of intersection of the image curves through Q .



[4.3] Conformal mappings. The angle ϕ between the curves passing through q on the left is defined to be the angle between their tangents there. But as p and s merge with q , this angle $\phi \asymp \angle pqs$. Let the images of p , q , s be P , Q , S , respectively. Then to prove conformality, we need to show that $\angle PQS \asymp \phi \asymp \angle pqs$. To do so, we need to show that the transformation that carries the initial connecting vectors on the left to their “images” on the right (ultimately) does not alter the angle between them. This (linear) transformation is called the *Jacobian*, and its coordinate description in the case of a general mapping is derived via the next figure.

4.3.2 The Jacobian Matrix

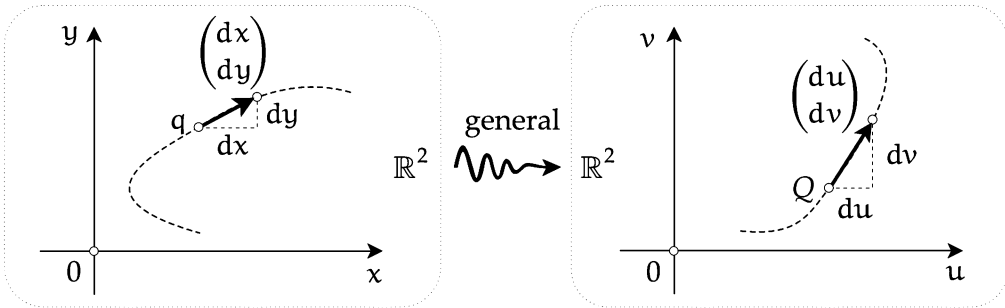
Consider [4.4]. As discussed, the direction of the illustrated curve through q is being described with an infinitesimal vector $\begin{pmatrix} dx \\ dy \end{pmatrix}$; the infinitesimal image vector $\begin{pmatrix} du \\ dv \end{pmatrix}$ gives the direction of the image curve through Q . We can determine the component du of $\begin{pmatrix} du \\ dv \end{pmatrix}$ as follows:

$$\begin{aligned}
 du &= \text{total change in } u \text{ due to moving along } \begin{pmatrix} dx \\ dy \end{pmatrix} \\
 &= (\text{change in } u \text{ produced by moving } dx \text{ in the } x\text{-direction}) \\
 &\quad + \\
 &\quad (\text{change in } u \text{ produced by moving } dy \text{ in the } y\text{-direction}) \\
 &= (\text{rate of change of } u \text{ with } x) \cdot (\text{change } dx \text{ in } x) \\
 &\quad + \\
 &\quad (\text{rate of change of } u \text{ with } y) \cdot (\text{change } dy \text{ in } y) \\
 &= (\partial_x u) dx + (\partial_y u) dy,
 \end{aligned}$$

where $\partial_x = \partial/\partial x$ etc. Likewise, we find that the vertical component is given by the formula

$$dv = (\partial_x v) dx + (\partial_y v) dy.$$

Since these expressions are linear in dx and dy , it follows (assuming that not all the partial derivatives vanish) that the infinitesimal vectors are carried to their images by a *linear transformation*. The general significance of this will be discussed later, but for the moment it means that the local effect of our mapping is completely described by a matrix J called the *Jacobian*. Thus,



[4.4] The local linear (Jacobian) transformation induced by a general mapping. An infinitesimal vector along a curve is carried to its image along the image curve by a linear transformation called the *Jacobian*, J . As the text explains, $du = (\partial_x u) dx + (\partial_y u) dy$, and similarly for dv , leading to the matrix representation of J as (4.2).

$$\begin{pmatrix} dx \\ dy \end{pmatrix} \mapsto \begin{pmatrix} du \\ dv \end{pmatrix} = J \begin{pmatrix} dx \\ dy \end{pmatrix},$$

where the Jacobian matrix is

$$J = \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix}. \quad (4.2)$$

We are now in a position to return to the specific mapping $z \mapsto z^2$, or more precisely to the mapping of $\mathbb{R}^2$ that we extracted from it. If we evaluate (4.2) for the mapping (4.1), we find that

$$J = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix}.$$

The geometric effect of this matrix is perhaps more clearly seen if we switch to polar coordinates. At the point $z = r e^{i\theta}$ —or rather $(r \cos \theta, r \sin \theta)$, since for the moment we are still in $\mathbb{R}^2$ —we have

$$J = 2r \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

The effect of the $2r$ is merely to expand all the vectors by this factor. This clearly does not affect the angle between any two of them. The remaining matrix is probably familiar to you as producing a rotation of θ , and hence it too does not alter the angle between vectors. Since both stages of the transformation preserve angles, we have in fact verified the previous claim: the net transformation is conformal.

4.3.3 The Amplitwist Concept

We have just seen that the local effect of $z \mapsto z^2$ on infinitesimal vectors is to expand them and to rotate them. Transformations of this type (i.e., whose local effect is produced in these two steps) will play a dominating role from now on, and it will be very much to our advantage to have vivid new words specifically to describe them.

If all the infinitesimal vectors ($\overrightarrow{qp}$ etc.) emanating from q merely undergo an equal enlargement to produce their images at Q , then we shall say that the local effect of the mapping is to *amplify* the vectors, and that the magnification factor involved is *the amplification* of the mapping at the point q . If, on the other hand, they all undergo an equal rotation, then we shall say that the local effect of the mapping is to *twist* the vectors, and that the angle of rotation involved is *the twist* of the mapping at the point q . More generally, the kind of mapping that will concern us will locally both amplify *and* twist infinitesimal vectors—we say that such a transformation is locally *an amplitwist*. Thus “an amplitwist” is synonymous

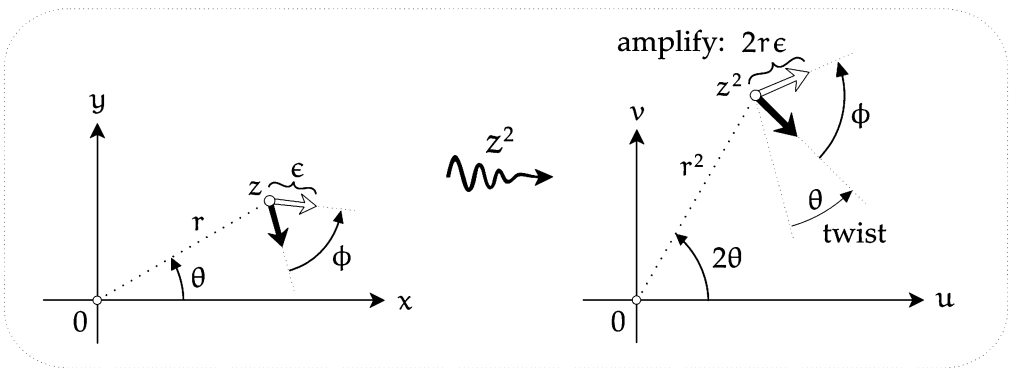
with “a (direct) similarity”, *except* that the former refers to the transformation of *infinitesimal* vectors, whereas “a similarity” has no such connotation.

[We remind the reader of the discussion in both Prefaces: “infinitesimal” is being used here in a definite, technical sense—small and ultimately vanishing, the relationship to other infinitesimals being expressed via Newtonian “ultimate equalities”.]

We can illustrate the new terminology with reference to the concrete case we have just analysed: The mapping $z \mapsto z^2$ is locally an amplitwist with amplification $2r$ and twist θ . See [4.5]. Quite generally, this figure makes it clear that *if a mapping is locally an amplitwist then it is automatically conformal*—the angle ϕ between the infinitesimal complex numbers is preserved.

Returning to [4.1] and [4.2], we now understand why infinitesimal squares were mapped to infinitesimal squares. Indeed, an infinitesimal region of *arbitrary* shape located at z will be “amplitwisted” (amplified and twisted) to a similar shape at z^2 . Note that here we are extending our terminology still further: henceforth we will freely employ the verb “to amplitwist”, meaning to amplify and to twist an infinitesimal geometric object.

All we really have at the moment is one simple mapping that turned out to be locally an amplitwist. In order to appreciate how truly fundamental this amplitwist concept is, we must return to $\mathbb{C}$ and begin from scratch to develop the idea of complex differentiation.



[4.5] Amplitwist terminology. The local effect of $z \mapsto z^2$ on infinitesimal complex numbers emanating from $z = r e^{i\theta}$ is to amplitwist them. Specifically, here we must amplify them by $2r$ —this magnification factor is called the amplification—and we must twist them by θ —this rotation angle is called the twist. The net transformation is called the amplitwist: it is equivalent to multiplication by (amplification) $e^{i(\text{twist})} = 2r e^{i\theta}$, in this example. If a complex mapping is locally an amplitwist, as this one is, then it is also conformal, as illustrated. We reserve this terminology for infinitesimal complex numbers, i.e., for small complex numbers in their moment of vanishing.

4.4 The Complex Derivative as Amplitwist

4.4.1 The Real Derivative Re-examined

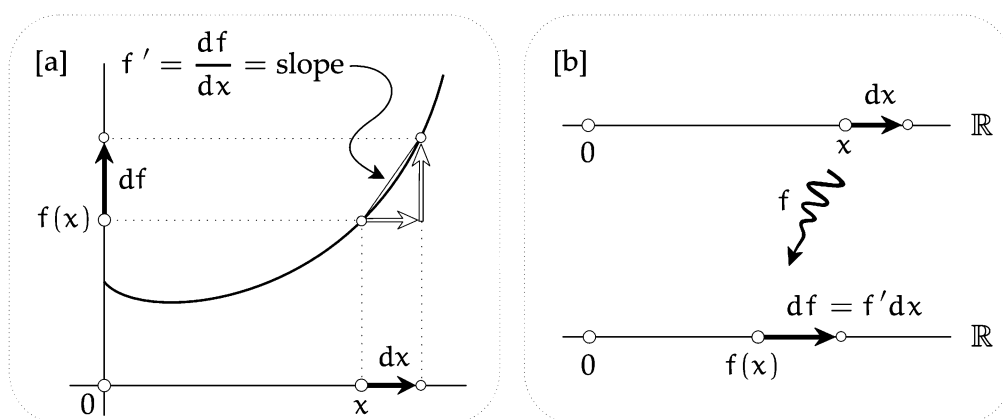
In the ordinary real calculus we have a potent means of visualizing the derivative f' of a function f from $\mathbb{R}$ to $\mathbb{R}$, namely, as the slope of the graph $y = f(x)$. See [4.6a]. Unfortunately, due to our lack of four-dimensional imagination, we can't draw the graph of a complex function, and hence we cannot generalize this particular conception of the derivative in any obvious way.

As a first step towards a successful generalization, we simply split the axes apart, so that [4.6a] becomes [4.6b]. Note that we have drawn both copies of $\mathbb{R}$ in a horizontal position, in anticipation of their being viewed as merely the real axes of two complex planes.

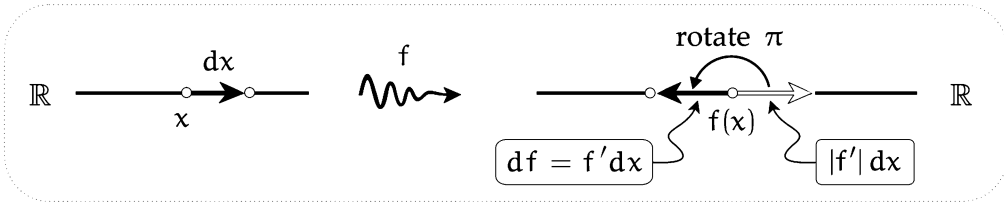
Next, continuing in the spirit of the previous section, we observe that $|f'(x)|$ describes how much the initial infinitesimal vector at x must be expanded to obtain its image at $f(x)$. More algebraically, $f'(x)$ is that real number by which we must multiply the initial vector to obtain its image:

$$f'(x) \cdot \rightarrow = \longrightarrow. \quad (4.3)$$

If $f'(x) > 0$ (as in [4.6b]) then the image of positive dx is a positive df , but if $f'(x) < 0$ then the infinitesimal image vector df is negative and points to the left, as illustrated in [4.7]. In this case, df can be obtained by first expanding dx by $|f'(x)|$, then *rotating it by π* . If we think of $f'(x)$ as a point on the real axis of $\mathbb{C}$, then $\arg[f'(x)] = 0$ when $f'(x) > 0$, and $\arg[f'(x)] = \pi$ when $f'(x) < 0$. Thus,



[4.6] The real derivative re-examined. [a] The traditional picture of the derivative f' as the slope of the tangent to the graph $y = f(x)$. [b] Let us instead break apart the two copies of the real axis, as we are forced to break apart the two copies of $\mathbb{C}$ in the case of a complex function, $z \mapsto f(z)$. Then an infinitesimal movement dx is mapped to $df = f' dx$: in other words, the local transformation is an amplification by f' .



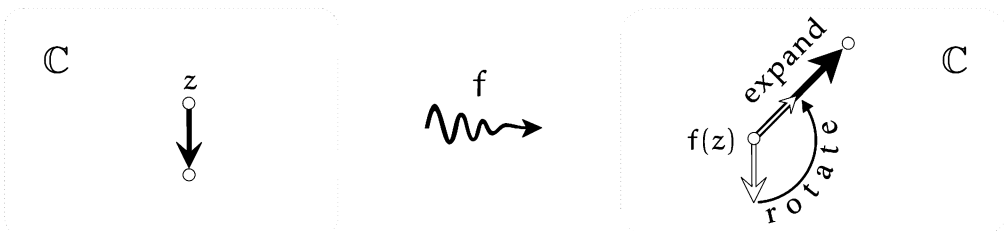
[4.7] If we think of $f'(x)$ as a point on the real axis of $\mathbb{C}$, then $\arg[f'(x)] = 0$ when $f'(x) > 0$, and $\arg[f'(x)] = \pi$ when $f'(x) < 0$. Thus, regardless of whether $f'(x)$ is positive or negative, we see that the local effect of f on an infinitesimal vector dx at x is to expand it by $|f'(x)|$ and to rotate it by $\arg[f'(x)]$.

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With all this fresh in our minds, we now attempt to generalize the notion of “derivative” to mappings of $\mathbb{C}$.

4.4.2 The Complex Derivative

Consider the effect of a complex mapping $f(z)$ on an infinitesimal complex number emanating from z . Its image (i.e., the connecting complex number between the two image points) will be an infinitesimal complex number emanating from $f(z)$. The generalization of [4.6b] or [4.7] is now [4.8]. On the right, we have drawn this image complex number in black, and we have also drawn a copy [white] at $f(z)$ of the original arrow at z . To transform the white arrow into the black image arrow now requires not only an expansion, but also a rotation. In figure [4.8] it looks as though we must expand the white arrow by 2, and rotate it by $(3\pi/4)$. Contrast this with the case of a real function, where the required rotation angle could only be 0 or π ; in the case of a complex function we need rotations through arbitrary angles.



[4.8] The local effect of a general complex mapping. The effect of a complex function on an infinitesimal arrow is to rotate it and expand it, which is equivalent to multiplication by a complex number $= (\text{expansion factor})e^{i(\text{rotation angle})} = 2e^{i(3\pi/4)}$, in the illustrated example.

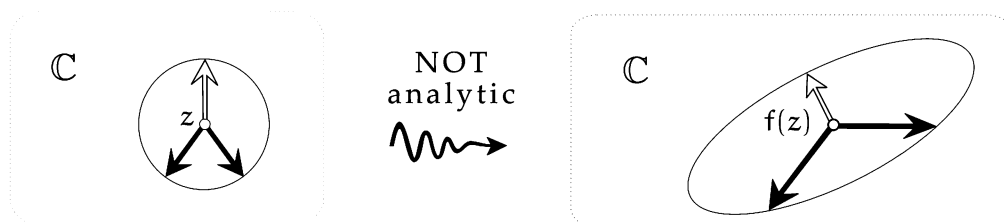
Nevertheless, we can still write down an algebraic equation completely analogous to (4.3), because “*expand and rotate*” is *precisely what multiplication by a complex number means*. Thus the complex derivative $f'(z)$ can now be introduced as that complex number by which we must multiply the infinitesimal number at z to obtain its image at $f(z)$:

$$f'(z) \cdot \downarrow = \nearrow. \quad (4.4)$$

In order to produce the correct effect, *the length of $f'(z)$ must be the magnification factor, and the argument of $f'(z)$ must be the angle of rotation*. For example, at the particular point shown in [4.8] we would have $f'(z) = 2e^{i(3\pi/4)}$. In fact, in the spirit of Chapter 1, we need not even distinguish between the local transformation and the complex number that represents it.

To find $f'(z)$ we have looked at the image of a specific arrow at z , but (unlike the case of $\mathbb{R}$) there are now infinitely many possible directions for such arrows. What if we had looked at an arrow in a different direction from the illustrated one?

We are immediately in trouble, because a typical mapping¹ will do what you see in [4.9]. Clearly the magnification factor differs for the various arrows, and likewise each arrow needs to be rotated a different amount to obtain its image. While we could still use a complex number in (4.4) to describe the transformation of the arrows, it would have to be a *different* number for each arrow. There would therefore be no single complex number we could assign to this point as being *the* derivative of f at z . We have arrived at an apparently gloomy impasse: a typical mapping of $\mathbb{C}$ simply cannot be differentiated.



[4.9] Infinitesimal complex numbers in different directions are generally rotated and expanded by different amounts. *In general, a complex function will map an infinitesimal circle to an infinitesimal ellipse, as shown. But this means that infinitesimal radii in different directions each undergo different rotations and expansions. The mapping is not locally an amplitwist, and is therefore not “analytic”: $f'(z)$ does not exist.*

¹ Shortly we will justify certain details of [4.9], such as the fact that an infinitesimal circle is mapped to an infinitesimal ellipse.

4.4.3 Analytic Functions

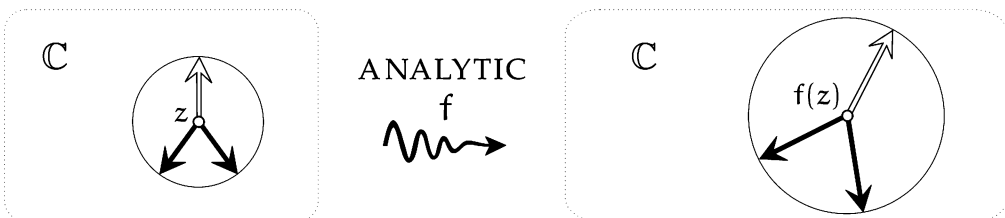
We get around the above obstacle in Zen-like fashion—by ignoring it! That is, from now on we concentrate almost exclusively on those very special mappings that *can* be differentiated. Such functions are called *analytic* (or *holomorphic*). From the previous discussion it follows that

Analytic mappings are precisely those whose local effect is an amplitwist: all the infinitesimal complex numbers emanating from a single point are amplified and twisted the same amount.

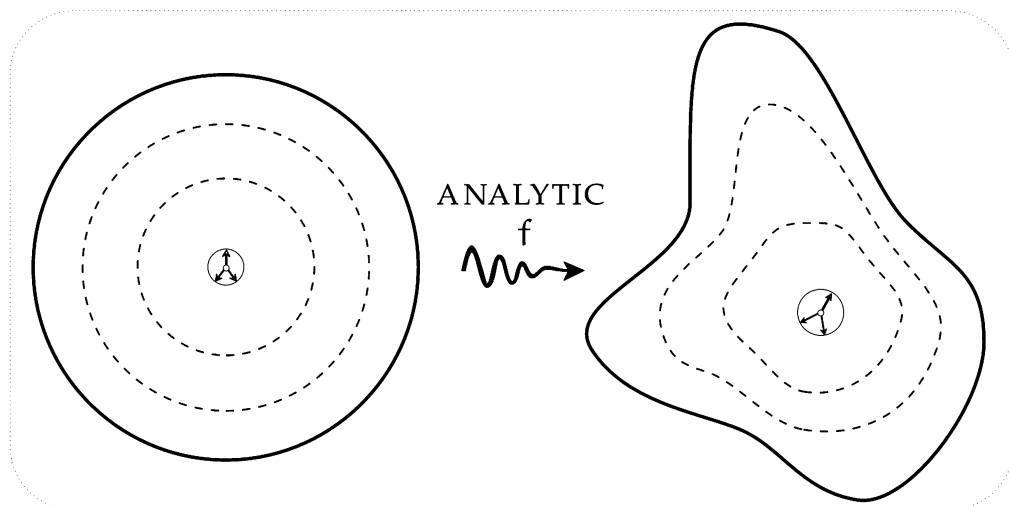
In contrast to [4.9], the effect of an analytic mapping can be seen in [4.10]. For such a mapping the derivative exists, and simply *is* the amplitwist, or, if you prefer, the complex number representing the amplitwist.

At this point you might quite reasonably fear that however interesting such mappings might be, they would be too exotic to include any familiar or useful functions. However, a ray of hope is held out by the humble-looking mapping $z \mapsto z^2$, for we have already established that it is locally an amplitwist, as illustrated in [4.5], and so it now gains admittance into the select set of analytic functions. In fact, quite amazingly, we will discover in the next chapter that virtually *every* function we have met in this book is analytic! Of course we have already seen plenty of empirical evidence of this in our many pictures showing small “squares” being mapped to small “squares”.

It should perhaps be stressed that all our recent pictures have been concerned with local properties, and hence with *infinitesimal* arrows and figures. For example, it’s clear from [4.10] that any analytic mapping will send infinitesimal circles to other infinitesimal circles; however, this does *not* mean that such mappings typically send circles to circles. Figure [4.11] (which contains [4.10] at its centre) illustrates the fact that if we start with an infinitesimal circle and then expand it, its image will generally distort out of all semblance of circularity. Of course, an important exception to this is provided by the Möbius transformations, for these



[4.10] Analytic = locally an amplitwist, expanding and rotating all infinitesimal complex numbers equally. By definition, an analytic mapping is locally an amplitwist, and the derivative $f'(z)$ simply is the amplitwist.



[4.11] Analytic mappings only send infinitesimal circles to circles. While an analytic mapping must send infinitesimal circles to circles, it will generally deform an expanding circle into a shape that becomes less and less circular. The only analytic mappings that preserve circles of all sizes are the Möbius transformations.

precisely do preserve circles of all sizes. In fact it can be shown that the Möbius transformations are the *only*² ones with this property.

4.4.4 A Brief Summary

The principal kinds of mapping we wish to study in this book are the analytic (complex-differentiable) ones. Although these will turn out to include almost all the useful functions, they are nevertheless very special. Their effect on an infinitesimal disc centred at z is, after translation to $f(z)$, simply to amplify and twist it. The *amplification* is the expansion factor, and the *twist* is the angle of rotation. The local effect of f is then completely encoded in the single complex number $f'(z)$, the derivative of f , or (as we will often prefer to call it) the *amplitwist* of f :

$$\begin{aligned}
 f'(z) &= \text{the derivative of } f \text{ at } z \\
 &= \text{the amplitwist of } f \text{ at } z \\
 &= (\text{amplification}) e^{i(\text{twist})} \\
 &= |f'(z)| e^{i \arg[f'(z)]}.
 \end{aligned}$$

To obtain the image at $f(z)$ of an infinitesimal complex number at z , you just multiply it by $f'(z)$.

Two last points. We have introduced the word “amplitwist” (in addition to “derivative”) because it is suggestive, and because it will make later reasoning

² This was proved by Carathéodory (1937).

much easier to explain. However, the student meeting this subject for the first time should be made aware of the fact that in all other books *only* the word derivative is used. Also, note that the two words are synonymous only to the extent (cf. Chapter 1) that a complex number can be identified with the similarity transformation it produces when each point is multiplied by it. Thus, for example, “to differentiate” will *not* mean the same as “to amplitwist”: the former refers to the act of finding the derivative of a function, while the latter refers to the act of “amplifying and twisting” an infinitesimal geometric figure.

4.5 Some Simple Examples

In the following examples we have superimposed the image copy of $\mathbb{C}$ on the original one.

$$z \mapsto z + c.$$

This represents a translation of the points by c . As we see in [4.12a], the length of complex numbers emanating from z is preserved, and hence the amplification is unity. Equally clearly, since no rotation is induced, the twist is zero. Hence

$$(z + c)' = \text{amplitwist of } (z + c) = 1 e^{i0} = 1.$$

Notice how this is in complete accord with the familiar rule of real calculus, namely, that $\frac{d}{dx}(x + c) = 1$.

$$z \mapsto Az.$$

If $A = a e^{i\alpha}$, then this represents the combination of an origin-centred expansion by a , and a rotation by α . It is clear in [4.12b] that any arrow at z (in particular an infinitesimal one) will suffer precisely the same amplification and twist as do the points of the plane themselves. Hence

$$(Az)' = \text{amplitwist of } (Az) = A.$$

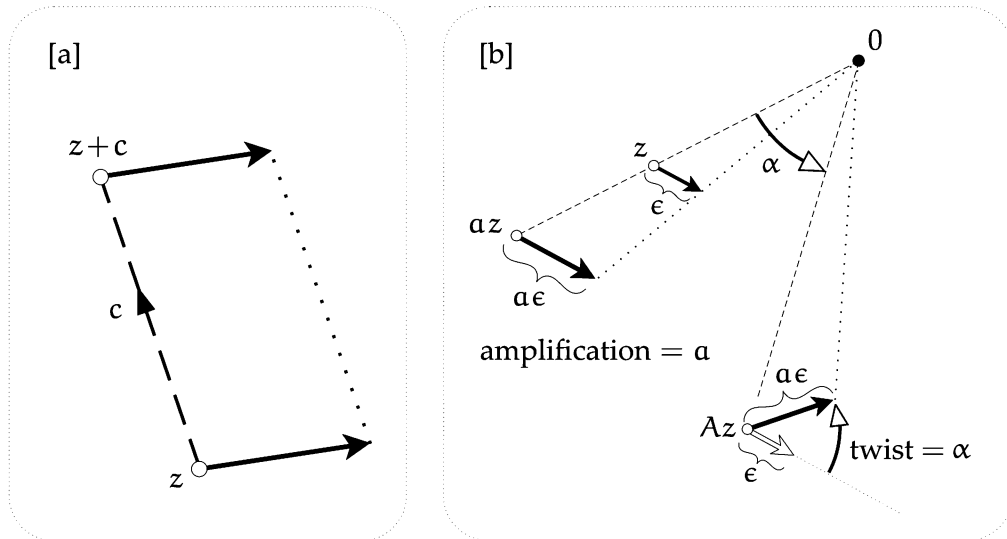
While the meaning is richer, this is once again formally identical to the familiar result $\frac{d}{dx}(Ax) = A$.

$$z \mapsto z^2.$$

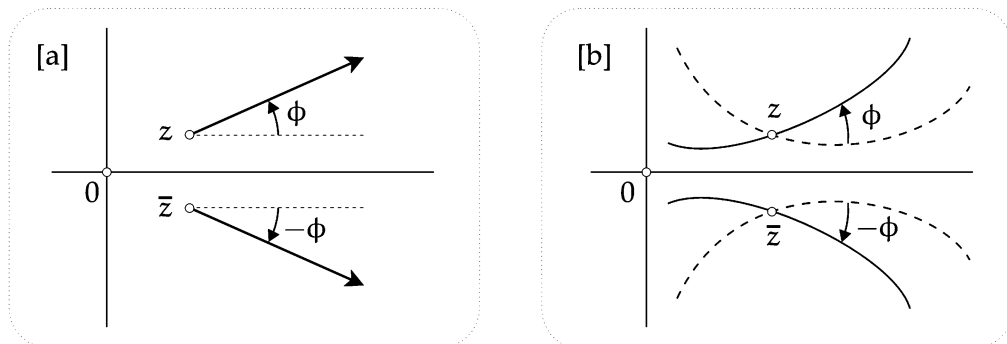
Our earlier investigation revealed that at the point $z = r e^{i\theta}$ this mapping is locally an amplitwist with amplification $2r$ and twist θ , as illustrated in [4.5]. Hence,

$$(z^2)' = \text{amplitwist of } (z^2) = (\text{amplification}) e^{i(\text{twist})} = 2r e^{i\theta} = 2z.$$

Once again, note that this result is formally identical to the formula $(x^2)' = 2x$ of ordinary calculus. In the next chapter we shall obtain a directly complex and geometrical demonstration of this fact.



[4.12] The complex amplitwist formula agrees with the real derivative. [a] The translation $z \mapsto z + c$ has amplification 1 and twist 0, so its amplitwist is 1, in accordance with the real formula, $(x+c)' = 1$. [b] The mapping $z \mapsto Az = a e^{i\alpha} z$ has amplification a and twist α , so its amplitwist is $a e^{i\alpha} = A$, in accordance with the real formula, $(Ax)' = A$.



[4.13] Conjugation is anticonformal and therefore not analytic. The mapping $z \mapsto \bar{z}$ is anticonformal, so it is not locally an amplitwist, and the derivative does not exist.

$$z \mapsto \bar{z}.$$

Since this mapping is anticonformal, it clearly cannot be analytic, for we have already observed that if a mapping is locally an amplitwist, then it is automatically conformal. Figure [4.13a] pinpoints the trouble. From the picture we see that the image at $\bar{z}$ of any complex number emanating from z has the same length as the original, and hence the amplification is unity. The problem lies in the fact that an arrow at angle ϕ must be rotated by -2ϕ to obtain its image arrow at angle $-\phi$. Thus different arrows must be rotated different amounts (which is not a twist) and hence there is no amplitwist.

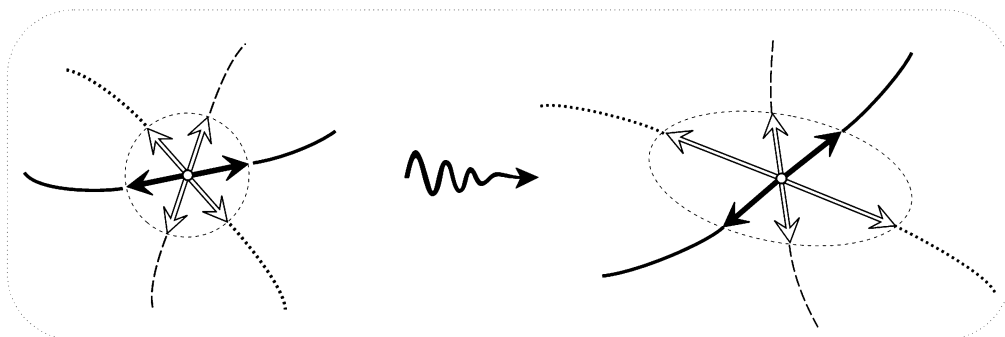
4.6 Conformal = Analytic

4.6.1 Introduction

In [4.5] we saw clearly that any mapping that is locally an amplitwist is also automatically conformal. In terms of complex differentiation, we can now rephrase this by saying that all analytic functions are conformal. The question then naturally arises as to whether the converse might also be true. Is every conformal mapping analytic, or, in other words, is the local effect of every conformal mapping nothing more complicated than an amplitwist? If this were the case then the two concepts would be equivalent and we would have a new way of recognizing, and perhaps reasoning about, analytic functions. A tempting prospect!

To dismiss this as a possibility would only require the discovery of a single function that is conformal and yet whose local effect is not an amplitwist. The example of complex conjugation, illustrated in [4.13b], shows how important it is that we take into account the fact that the mapping preserves not only the magnitude of angles, but also their *sense*. For $z \mapsto \bar{z}$ is not analytic, but it is also not a counterexample to the conjecture, because it is *anticonformal*.

We have seen that although conjugation does possess an amplification, it fails to be analytic because it doesn't have a twist. Let us now consider instead a function that does possess a twist, but which again fails to be analytic, this time by virtue of not having an amplification. The effect of such a mapping at a particular point is illustrated in [4.14]. The three curves on the LHS intersect at equal angles of $(\pi/3)$, and on the RHS their images do too. But the picture clearly shows that we are not dealing with an amplitwist. Imagine that the infinitesimal tangent complex numbers to the curves are first twisted, but then rather than being amplified, as they would be by an analytic function, they are expanded by *different* factors. Despite



[4.14] A twist without an amplitwist. At a single point it is possible for a mapping to possess a twist, and therefore to be conformal at that point, and yet not possess an amplification, and therefore not be an amplitwist.

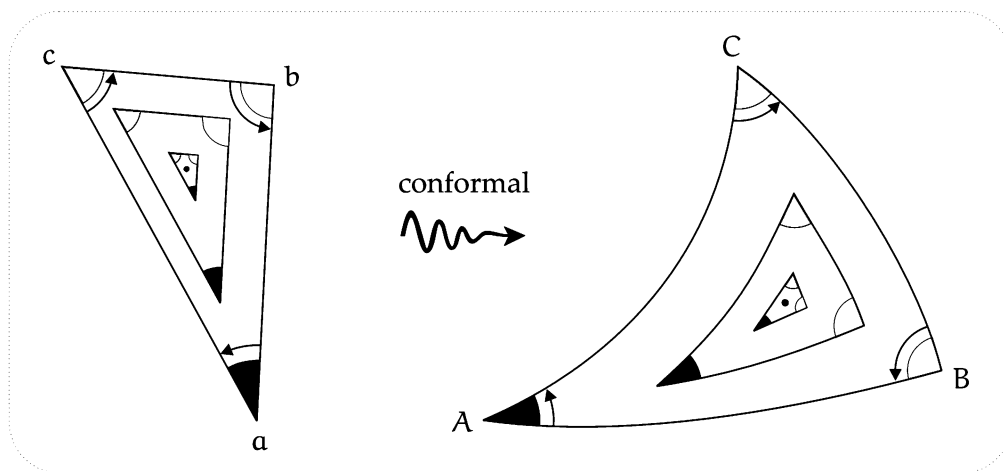
this, however, the initial twist ensures that the angle between two curves is preserved both in magnitude and sense: the mapping is genuinely conformal at this point.

4.6.2 Conformality Throughout a Region

If we only insist on conformality at isolated points then such counterexamples do indeed exist (we've drawn one!), but if we require the mapping to be conformal *throughout a region* then this nonanalytic behaviour cannot occur.

Imagine that we have a region throughout which the mapping is (i) conformal, and (ii) sufficiently non-pathological that an infinitesimal line-segment is mapped to another infinitesimal line-segment. In fact, re-examination of [4.3] reveals that (ii) must be presupposed in order for (i) even to make sense. For if the infinitesimal straight piece of curve from q to p did not map to another of the same kind at Q , then we could not even speak of an angle of intersection at Q , let alone of its possible equality with ϕ .

Now look at [4.15]. In our conformal region we have drawn a large (i.e., not infinitesimal) triangle abc , along with its image ABC . Notice that while the straight edges of abc are completely distorted to produce the curvilinear edges of ABC , the angles of this "triangular" image are identical with those of the original. Now imagine shrinking abc down towards an arbitrary point in the region. As we do so, the sides of its shrinking image will increasingly resemble straight lines [by virtue



[4.15] A mapping that is conformal throughout a region must locally be an amplitwist. If the mapping is conformal through the region in which the triangle abc shrinks to a point, then its curvilinear triangular image ABC must ultimately become a similar infinitesimal triangle, which can therefore be obtained by amplitwisting the original infinitesimal triangle.

of (ii)], and all the while the angles will remain the same as the original's. Thus, any infinitesimal triangle in this region is mapped to another infinitesimal *similar* triangle. Since the image triangle merely has a different size and alignment on the page, it is indeed obtained by amplitwisting the original.

We have thus established the sought-after equivalence of conformal and analytic mappings:

A mapping is locally an amplitwist at a point p if it is conformal throughout an infinitesimal neighbourhood of p .

For this reason, the conventional definition of f being “analytic” at p is that f' exist at p and at all points in an infinitesimal neighbourhood of p .

From this result we can immediately deduce, for example, that complex inversion $z \mapsto (1/z)$ is analytic, for we have already demonstrated geometrically that it is conformal. By the same token, it follows more generally that all Möbius transformations are analytic.

For no extra charge, we can obtain a further equivalence simply by concentrating on distances rather than angles. What we have just seen is that a mapping cannot possess a twist throughout a region without also having an amplification. In order to investigate the converse, suppose that a mapping is only known to possess an amplification throughout a region. Re-examine [4.15] from this point of view. Unlike the previous case, there is no longer any *a priori* reason for the image ABC to betray any features common to the original. However, as we carry out the same shrinking process as before, the local existence of amplifications begins to reveal itself.

As the triangle becomes very small, we may consider two of its sides, for example ab and ac , to be infinitesimal arrows emanating from a vertex. While we may not yet know anything of angles, we do know that these arrows both undergo the same amplification to produce their images AB and AC . But if we now apply this reasoning at one of the other vertices, we immediately find that in order to be consistent, *all three* sides must undergo the same³ amplification. Once again we have been able to deduce that the image triangle is similar to the original.

However, this time all we know is that the *magnitude* of the angles in the infinitesimal image triangle are the same as those in the original. If the sense of the angles also agree, then the image is obtained by amplitwisting the original, just as before. But if the angles are reversed, then we must flip the original triangle over as well

³ We only mean “same” in the sense that the variations in amplification are of the same infinitesimal order as the dimensions of abc . If the amplifications were *precisely* the same, then extending our argument to a whole network of closely spaced vertices, we would conclude that the amplification was constant throughout the region.

as amplitwisting it. This “flip” may be accomplished by reflecting in any line; in particular, we may employ reflection in the real axis, $z \mapsto \bar{z}$. Thus if $f(z)$ is a mapping that is known to possess an amplification throughout an infinitesimal neighbourhood of a point p , then either $f(z)$ is analytic at p , or else $\overline{f(z)}$ is analytic at p .

It is interesting to note that the use of triangles in the above arguments was not incidental, but instead crucial. Rectangles, for example, would simply not have sufficed. Take the first argument. Certainly conformality still guarantees us that an infinitesimal rectangle maps to another infinitesimal rectangle. However, this image rectangle could in principle have very different proportions from the original, and hence not be obtainable via an amplitwist. Try the second argument for yourself and see how it too fails.

For a computational approach to the above results, see Ahlfors (1979, p. 73).

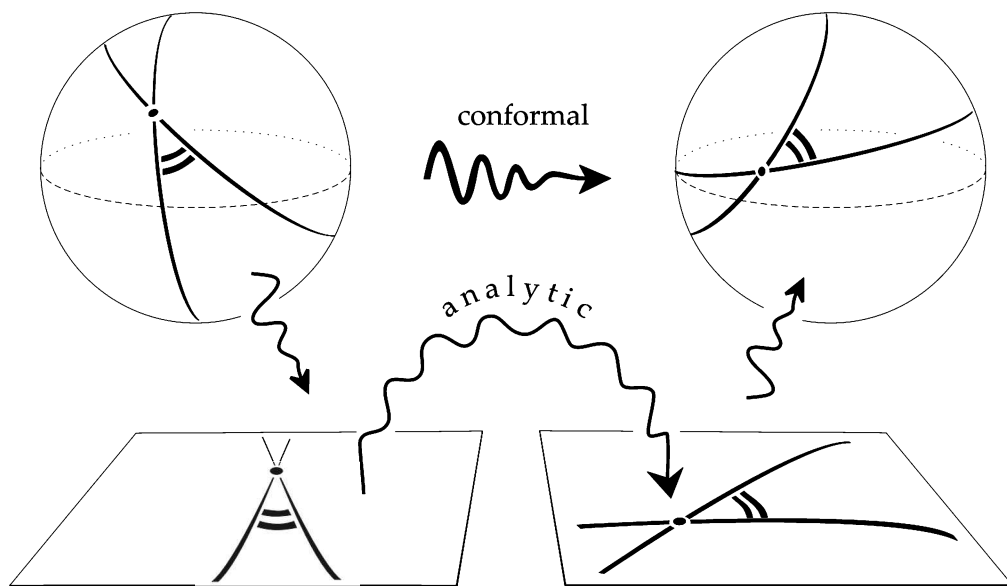
4.6.3 Conformality and the Riemann Sphere

In the previous chapter we addressed a twin question: “How are we to visualize the effect of a mapping on infinitely remote parts of the complex plane, or the effect of a mapping that hurls finite points into the infinite distance?” Our answer was to replace both complex planes (original and image) with Riemann spheres. We could then visualize the mapping as taking place between the two spheres, rather than between the two planes. To a large extent the success of this merely depended upon the fact that we had gathered up the infinite reaches of the plane to a single point on the sphere. It did not depend on the precise manner in which we chose to do this. Why then the insistence on accomplishing this with stereographic projection, rather than in some other way? Several reasons emerged in the previous chapter, but the present discussion shows that another compelling reason is that stereographic projection is *conformal*.

Only now can we fully appreciate this point, for we have seen that analytic functions are the conformal mappings of the plane. As illustrated in [4.16], the conformality of stereographic projection now enables us to translate this directly into a statement about Riemann spheres:

A mapping between spheres represents an analytic function if and only if it is conformal.

We have drawn the spheres separate from the planes to reinforce the idea that we are entitled to let the plane fade from our minds, and to adopt instead the sphere as a logically independent base of operations. Indeed, at this stage we could consider complex analysis to be nothing more than the study of conformal maps between spheres. But in works on Riemann surfaces it is shown that in order to embrace the global aspects of many-one functions and their inverses, one must extend this conception to conformal mappings between more general surfaces, such as doughnuts.



[4.16] Analytic mappings of $\mathbb{C}$ induce conformal mappings of the Riemann sphere, for the simple reason that stereographic projection is conformal.

4.7 Critical Points

4.7.1 Degrees of Crushing

We return to the mapping z^2 and note that at $z = 0$, $(z^2)' = 2z = 0$. A place such as this, where the derivative vanishes, is called a *critical point*. Recall that in the previous chapter we defined the term “critical point” differently, as a point at which the conformality of an otherwise conformal mapping breaks down. These two definitions are not at odds with one another. If the derivative $f'(z)$ of an analytic mapping f is not zero at $z = p$, then we know that f is conformal at p , so conformality can only break down at points where $f'(z) = 0$. Although it is not obvious, later we will be able to prove the converse fact that if $f'(p) = 0$ then f *cannot* be conformal at p . Thus the two definitions are equivalent.

In terms of the ampliwist concept a critical point could equally well be defined as a point of zero amplification. This suggests that the effect of an analytic mapping on an infinitesimal disc centred at a critical point is to “crush it down to a single image point”. The statement in quotes is not to be taken literally, rather it is to be understood in the following sense.

Imagine that the disc (radius ϵ) is so tiny that it must be placed under a microscope in order to be seen. Suppose that we have available a whole family of lenses of increasing power with which to view it: $L_0, L_1, L_2, L_3, \dots$. For example, L_0 has magnification $1/\epsilon^0 = 1$, so that it's really no better than using the naked eye. On the other hand L_1 has magnification $1/\epsilon^1$, and it is thus so powerful that we can

actually see the disc with it. The lens L_2 is even more remarkable in that it magnifies by $1/\epsilon^2$, so that even a small part of our microscopic disc now completely fills the viewing screen⁴.

Let's switch back to L_1 so that we can see the whole disc again, and watch what happens to it when we apply the transformation $z \mapsto z^2$. It disappears! At best we might see a single dot sitting at the image of the critical point. It is in *this* sense that the mapping is crushing. However, if we now attach L_2 instead, we can see our mistake: the dot isn't a dot, in fact it's another disc of radius ϵ^2 .

For this particular mapping, L_2 was sufficient to see that the disc had not been completely crushed. However, at a critical point of another mapping, even this might not provide sufficient magnification, and we would require a stronger lens, say L_m , to reveal that the image of the disc isn't just a point. The integer m measures the degree of crushing at the critical point.

4.7.2 Breakdown of Conformality

In addition to being locally crushing, we have stated (but not yet proved) that the conformality of an analytic function breaks down at its critical points. We can see this in our example. When the z^2 mapping acts on a pair of rays through the critical point $z = 0$, it fails to preserve the angle between them; in fact it doubles it. Thus, just at the critical point, the conformality of z^2 breaks down. This is a general property. In fact we will show later that the behaviour of a mapping very near to a critical point is essentially given by z^m , $m \geq 2$. Rather than being conserved, angles at the critical point $z = 0$ are consequently multiplied by m . We quantify the degree⁵ of this strange behaviour by saying that $z = 0$ is a critical point of *order* $(m - 1)$. Notice that this m is the same one as in the previous paragraph: in order to see the image we have to use the L_m lens.

Despite the fact that conformality breaks down at critical points, we shall continue to make such bald statements as, " z^2 is conformal". The tacit assumption is that critical points are being excluded. Indeed we were making this assumption throughout the previous section, for we only concerned ourselves there with typical points. Later we will see that critical points are, in a mathematically precise sense, "few and far between", and this is our excuse for the scant attention we are presently paying them. Nevertheless, we may safely skirt around this issue

⁴ In terms of this analogy we could say that most of the diagrams in this chapter, indeed in the rest of the book, show views of the image complex plane taken through L_1 . For example [4.10] depicts a tiny circle being amplitwisted to produce another circle. However, if we viewed part of this image 'circle' with L_2 instead of L_1 , then deviations from circularity would become visible. Of course the smaller we make the preimage circle, the smaller these deviations will be.

⁵ The reason we define it to be $(m - 1)$, rather than m , is that this properly reflects the multiplicity of the root of the derivative.

only so long as we focus on the effect of the function on separate chunks of its domain. When one studies Riemann surfaces, one tries to fit all this partial information into a global picture of the mapping, and in achieving this the critical points will play a crucial role. They do so by virtue of yet another aspect of the peculiar behaviour of a mapping in the vicinity of such points, and it is to this feature that we now turn.

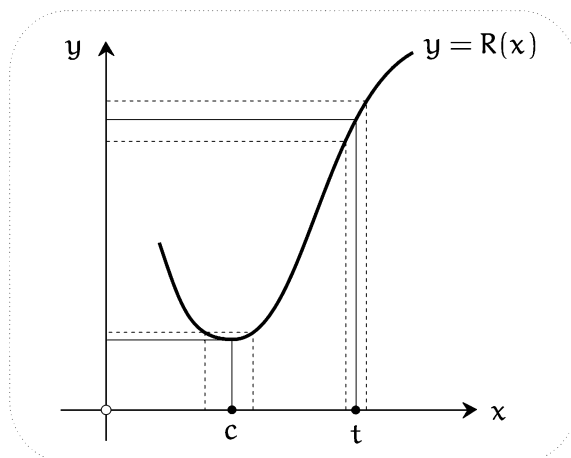
In the previous chapter we discussed the possibility of critical points being located at infinity. In particular, we considered $z \mapsto z^m$. On the Riemann sphere we drew two straight lines passing through the origin, and we thereby saw that angles at both $z = 0$ and $z = \infty$ were multiplied by m . We therefore conclude that ∞ is a critical point of z^m of order $(m - 1)$, just like the origin. Actually, except for $m = 2$, we don't yet know if z^m is conformal *anywhere*! However, in the next chapter we will see that it is conformal everywhere except at the two critical points we have just discussed.

4.7.3 Branch Points

First consider the case of a real function $R(x)$ from $\mathbb{R}$ to $\mathbb{R}$. In solving problems on maxima and minima, we learn from an early age the importance of finding the places where $R'(x) = 0$. Figure [4.17] shows an ordinary graph of $y = R(x)$, emphasizing a different aspect of the behaviour of R near to a "critical point" c where $R'(c) = 0$. Above a typical point t , for which $R'(t) \neq 0$, the graph is either going up or going down, so the function is locally one-to-one. However, near c it is clearly *two-to-one*.

An analogous significance holds for complex mappings. Typically $f'(z) \neq 0$, and so an infinitesimal neighbourhood of z is amplitwisted to an infinitesimal image neighbourhood of $w = f(z)$, and the two neighbourhoods are clearly in one-to-one correspondence. However, if $f'(z_0) = 0$ then (according to our earlier claim) near to z_0 the function behaves like z^m , with $m \geq 2$. Thus, if a point is in a close orbit around z_0 , its image will orbit w_0 at m -times the angular speed, and corresponding to each point near w_0 there will be m preimages near z_0 . Thus w_0 is a branch point of order $(m - 1)$. We conclude that *a critical point of a given order maps to a branch point of the same order*.

We began this idea by using an analogy with real functions, but we should also note an important difference. A real function $R(x)$ is necessarily one-to-one when $R'(x) \neq 0$, but (unlike the complex case) it *need not* be many-to-one when $R'(x) = 0$. The graph of x^3 , for example, is flat at the origin and yet it is still one-to-one in an infinitesimal neighbourhood of that point. In contrast to this, the complex mapping $z \mapsto z^3$ is three-to-one near the origin, due to the existence of complex cube roots.



[4.17] Breakdown of one-to-oneness near critical points. When $R'(x) \neq 0$ (as at $x = t$) the graph is either going up or down, so the function is one-to-one in the immediate vicinity of the point. But near a “critical point” $x = c$, defined by $R'(c) = 0$, we see that R need not be one-to-one. The same turns out to be true for complex analytic mappings, only more so: if $f'(c) = 0$, then f is necessarily not one-to-one in the immediate vicinity of $z = c$.

4.8 The Cauchy–Riemann Equations

4.8.1 Introduction

To end this chapter we will try to gain a better perspective on where the analytic functions lie within the hierarchy of mappings of the plane. A benefit of this will be the discovery of another way (the third!) of characterizing analytic functions, this time in terms of their real and imaginary parts.

The first thing to do is realize that the “general” mappings $(x, y) \mapsto (u, v)$ that we considered earlier were not really as general as they could have been. Picture part of the plane as being a rolled out piece of pastry on a table. A general mapping corresponds to “doing something” to the pastry, thereby moving its points to new locations (the images) on the table. For example, we might cut the pastry in half and move the two pieces away from each other. This is much more general than anything we contemplated earlier, for it does not even possess the rudimentary quality of *continuity*. That is, if two points are on either side of the cut, then no matter how close we move them together, their images will remain far apart.

Even if we do insist on continuity, the resulting mappings are still more general than those we have considered. For example, imagine pressing down the rolling-pin somewhere in the middle of the pastry, and, in a single roll, stretching the far side to twice its former size. This certainly is continuous, for bringing two points together always brings their images together. The problem now lies in the fact that if two infinitesimal, diametrically opposed arrows emanate from a point beneath

the starting position of the pin, then they each undergo a quite different transformation. Thus, in an obvious sense (not a subtle complex-differentiation sense) the mapping isn't differentiable at this point. Nevertheless, provided we stay away from this line, the mapping is differentiable in the real sense, and hence subject to our earlier analysis using the Jacobian matrix.

Another interesting kind of mapping arises from the commonplace operation of *folding* the pastry. Suppose we fold it like a letter being placed in an envelope [two creases]. Three different points will end up above a single point of the table, and the mapping is thus three-to-one. However, at the creases themselves the mapping is only one-to-one, and furthermore, differentiability also breaks down there. Nevertheless, provided that we only look at the fold-free portions of the pastry, such many-one functions are still subject to our previous analysis.

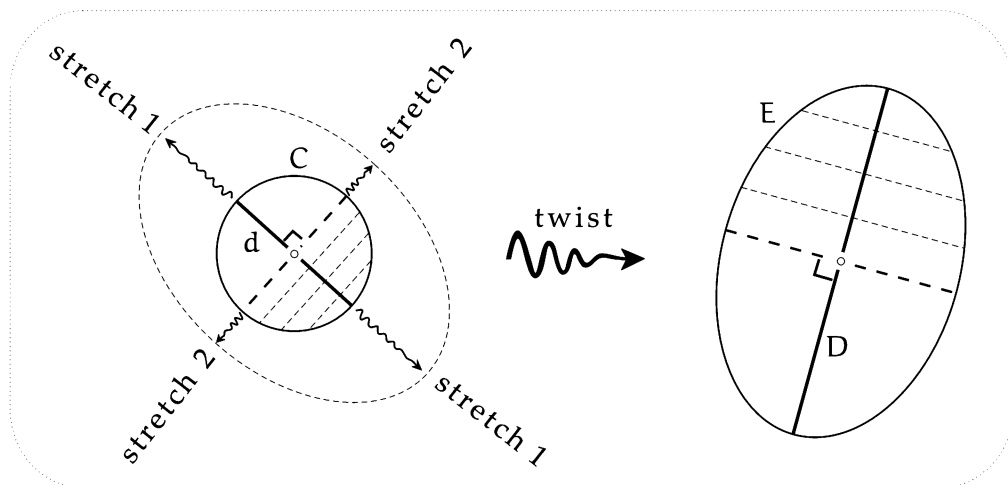
Suppose we play with the pastry in an ordinary way, rolling it (not necessarily evenly) now in this direction, now in another, then turning it, folding it, rolling it again, and so forth; then, provided we suitably restrict the domain, we can still apply our old analysis. While such a mapping is indeed very general, we hope that this discussion has revealed that (being continuous and differentiable in the real sense) it is, in fact, already quite high up the evolutionary ladder. It will therefore not come as such a surprise to learn that the local geometric effect of such a mapping is remarkably simple, though naturally not *as* simple as an amplitwist.

4.8.2 The Geometry of Linear Transformations

We pick up our earlier investigation where we left it. The local effect of the mapping is to perform the linear transformation encoded in the Jacobian matrix (4.2). If we can first understand the effect of a *uniform* linear transformation—corresponding to a constant matrix—then we shall be finished. For we need only then remember that our analysis is only applicable *locally*, the actual linear transformation varying as it does from one place to the next.

Consider the effect of a uniform linear transformation on a circle C . Since the Cartesian equation of C is quadratic, the linear change of coordinates induced by the transformation will lead to another quadratic equation for the image curve. The image curve E is thus a conic section, and since the finite points of C are not sent to infinity, this conic must be an *ellipse*. See [4.18], and compare this also with [4.9], where the local consequence of this result was illustrated for a *non*-uniform transformation acting on an infinitesimal circle.

We have just used an algebraic statement of linearity. The fundamental *geometric* fact is that it makes no difference if we add two vectors and then map the result, or if we map the vectors first and *then* add them. Convince yourself of these two simple consequences:



[4.18] Singular Value Decomposition. A linear transformation maps the circle C to the ellipse E . Let d be the diameter of C that is mapped to the major axis D . Consider the chords of C [dashed] that are perpendicular to d . Since these are all bisected by d , their images must be a family of parallel chords of E such that D is their common bisector. They must therefore be the family perpendicular to D . We thereby deduce the Singular Value Decomposition: A linear transformation is a stretch in the direction of d , another stretch perpendicular to it, and finally a twist.

- Parallel lines map to parallel lines.
- The midpoint of a line-segment maps to the midpoint of the image line-segment.

We now apply these facts to E .

Since all the diameters of C are bisected by the centre of C , it follows that the image chords of E must all pass through a common point of bisection. Thus the centre of C is mapped to the centre of E . Drawn in the same heavy line as its image is the particular diameter d of the circle that is mapped to the major axis D of the ellipse. Now consider the chords of C [dashed] that are perpendicular to d . Since these are all bisected by d , their images must be a family of parallel chords of E such that D is their common bisector. They must therefore be the family perpendicular to D . All this is summarized in [4.18].

Thus we have obtained the following result:

Local Singular Value Decomposition. The local linear transformation induced by a (non-linear) complex mapping is a stretch in the direction of d , another stretch perpendicular to it, and finally a twist. (4.5)

This result also makes sense at the level of counting degrees of freedom. Just as the matrix has four independent entries, so the specification of our transformation also requires four bits of information: the direction of d , the stretch factor in this direction, the perpendicular stretch factor, and the twist.

[REMARKS: This *Singular Value Decomposition* (SVD for short) is of enormous importance in Linear Algebra and its real-world applications. It was discovered by Eugenio Beltrami,⁶ whom we shall meet again in Chapter 6, because he also made pivotal discoveries in Hyperbolic Geometry. In the context of the standard SVD theorem, the two orthogonal expansion factors are called the *singular values* of the transformation. In VDGF⁷ we elaborate on these ideas, and use them to provide a *simple geometrical interpretation of the transpose of a matrix*, which we have not found in any standard Linear Algebra textbook. This in turn can be used to provide geometrical explanations of other standard results in Linear Algebra.]

The ultimate specialization to analytic functions now simply requires that the two stretch factors be put equal. This apparently reduces the number of degrees of freedom from four to three. However, since we are now producing an equal expansion in all directions, the direction chosen for d becomes irrelevant, and we are left with only *two* genuine degrees of freedom: the amplification and the twist.

Note that we now have the following:

An orientation preserving mapping is conformal if and only if it sends infinitesimal circles to infinitesimal circles.

If a mapping preserves circles in general, then, in particular, it must send infinitesimal circles to infinitesimal circles, and hence it must be conformal⁸. Bypassing the detailed investigation of the previous chapter, we now see that the conformality/analyticity of Möbius transformations follows from the mere fact that they preserve circles.

4.8.3 The Cauchy–Riemann Equations

We obtain another characterization of analytic functions if we now ask how we may recognize a Jacobian matrix for which both expansion factors are equal. This is most easily answered by considering what kind of matrix corresponds to multiplication by a complex number, for we already know that this produces the desired type of linear transformation. Multiplying $z = (x + iy)$ by $(a + ib)$, we get

$$(x + iy) \mapsto (a + ib)(x + iy) = (ax - by) + i(bx + ay).$$

This corresponds to the multiplication of a vector in $\mathbb{R}^2$ by the matrix

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}. \quad (4.6)$$

⁶ See Stewart (1993).

⁷ See Needham (2021, §15.4).

⁸ For a different proof of this fact, see Sommerville (1958, p. 237).

Compare this with the Jacobian matrix (4.2),

$$J = \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix}.$$

In order for the effect of J to reduce to an amplitwist, it must have the same form as (4.6), and thus

$$\begin{aligned} \partial_x u &= +\partial_y v, \\ \partial_x v &= -\partial_y u. \end{aligned} \tag{4.7}$$

These are the celebrated *Cauchy–Riemann equations*. They provide us with a third way of recognizing an analytic function. However, as with the underlying amplitwist concept, these equations must be satisfied throughout an infinitesimal neighbourhood of a point in order that the mapping be analytic there [see Ex. 12].

Since $(a + ib)$ is playing the role of the amplitwist, comparison of (4.6) and (4.2) now yields two formulae for the derivative:

$$f' = \partial_x u + i \partial_x v = \partial_x f, \tag{4.8}$$

and

$$f' = \partial_y v - i \partial_y u = -i \partial_y f. \tag{4.9}$$

By way of example, consider $z \mapsto z^3$. Multiplying this out we obtain a rather haphazard looking mess:

$$u + iv = (x^3 - 3xy^2) + i(3x^2y - y^3).$$

However, differentiating the real and imaginary parts, we obtain

$$\begin{aligned} \partial_x u &= 3x^2 - 3y^2 &= +\partial_y v, \\ \partial_x v &= 6xy &= -\partial_y u, \end{aligned}$$

and so the Cauchy–Riemann equations are satisfied. Thus, far from being haphazard, the special forms of u and v have ensured that the mapping is analytic. Using (4.8) we can calculate the amplitwist:

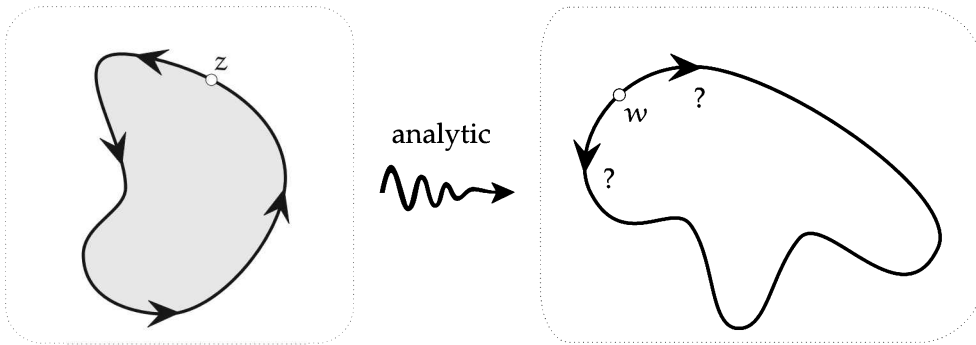
$$(z^3)' = 3(x^2 - y^2) + i 6xy = 3z^2,$$

just as in ordinary calculus. Check that (4.9) gives the same answer.

In the next chapter we will sever our umbilical cord to $\mathbb{R}^2$ and discover how the above results can be better understood by directly appealing to the geometry of the complex plane.

4.9 Exercises

- 1 Use the Cauchy–Riemann equations to verify that $z \mapsto \bar{z}$ is not analytic.
- 2 The mapping $z \mapsto z^3$ acts on an infinitesimal shape and the image is examined. It is found that the shape has been rotated by π , and its linear dimensions expanded by 12. Where was the shape originally located? [There are two possibilities.]
- 3 Consider $z \mapsto \Omega(z) = \bar{z}^2/z$. By writing z in polar form, find out the geometric effect of Ω . Using two colours, draw two very small arrows of equal length emanating from a typical point z : one parallel to z ; the other perpendicular to z . Draw their images emanating from $\Omega(z)$. Deduce that Ω fails to produce an amplitwist. [Your picture should show this in *two* ways.]
- 4 The picture shows the shaded interior of a curve being mapped by an analytic function to the exterior of the image curve. If z travels round the curve counter-clockwise, then which way does its image w travel round the image curve? [Hint: Draw some infinitesimal arrows emanating from z , including one in the direction of motion.]



- 5 Consider $f(x + iy) = (x^2 + y^2) + i(y/x)$. Find and sketch the curves that are mapped by f into (a) horizontal lines, and (b) vertical lines. Notice from your answers that f appears to be conformal. Show that it is *not* in two ways: (i) by explicitly finding some curves whose angle of intersection isn't preserved; and (ii) by using the Cauchy–Riemann equations.
- 6 Continuing from the previous exercise, show that *no* choice of v can make $f(x + iy) = (x^2 + y^2) + iv$ analytic.
- 7 (i) If $g(z) = 3 + 2i$ then explain geometrically why $g'(z) \equiv 0$.
 (ii) Show that if the amplification of an analytic function is identically zero (i.e., $f'(z) \equiv 0$) on some *connected* region, then the function is constant there.

- (iii) Give a simple counterexample to show that this conclusion does not follow if the region is instead made up of disconnected components.
- 8 Use pictures to explain why if $f(z)$ is analytic on some *connected* region, each of the following conditions forces it to reduce to a constant.
 - (i) $\operatorname{Re} f(z) = 0$
 - (ii) $|f(z)| = \text{const.}$
 - (iii) Not only is $f(z)$ analytic, but $\overline{f(z)}$ is too.
- 9 Use the Cauchy–Riemann equations to give rigorous computational proofs of the results of the previous two exercises.
- 10 Instead of writing a mapping in terms of its real and imaginary parts (i.e. $f = u + iv$), it is sometimes more convenient to write it in terms of length and angle:

$$f(z) = R e^{i\Psi},$$

where R and Ψ are functions of z . Show that the equations that characterize an analytic f are now

$$\partial_x R = R \partial_y \Psi \quad \text{and} \quad \partial_y R = -R \partial_x \Psi.$$

- 11 Let's agree to say that " $f = u + iv$ satisfies the Cauchy–Riemann equations" if u and v do. Show that if $f(z)$ and $g(z)$ both satisfy the Cauchy–Riemann equations, then their sum and their product do also.
- 12 For nonzero z , let $f(z) = f(x + iy) = xy/\bar{z}$.
 - (i) Show that $f(z)$ approaches 0 as z approaches any point on the real or imaginary axis, including the origin.
 - (ii) Having established that $f=0$ on both axes, deduce that the Cauchy–Riemann equations are satisfied at the origin.
 - (iii) Despite this, show that f is not even differentiable at 0, let alone analytic there! To do so, find the image of an infinitesimal arrow emanating from 0 and pointing in the direction $e^{i\phi}$. Deduce that while f does have a twist at 0, it fails to have an amplification there.
- 13 Verify that $z \mapsto e^z$ satisfies the Cauchy–Riemann equations, and find $(e^z)'$.
- 14 By sketching the image of an infinitesimal rectangle under an analytic mapping, deduce that the local magnification factor for area is the square of the amplification. Rederive this fact by looking at the determinant of the Jacobian matrix.
- 15 Let us define S to be the square region given by

$$a - b \leq \operatorname{Re}(z) \leq a + b \quad \text{and} \quad -b \leq \operatorname{Im}(z) \leq b.$$

- (i) Sketch a typical S for which $b < a$. Now sketch its image $\tilde{S}$ under the mapping $z \mapsto e^z$.
- (ii) Deduce the area of $\tilde{S}$ from your sketch, and write down the ratio

$$\Lambda \equiv \left(\frac{\text{area of } \tilde{S}}{\text{area of } S} \right).$$

- (iii) Using the results of the previous two exercises, what limit should Λ approach as b shrinks to nothing?
 - (iv) Find $\lim_{b \rightarrow 0} \Lambda$ from your expression in part (ii), and check that it agrees with your geometric answer in part (iii).
- 16 Consider the complex inversion mapping $I(z) = (1/z)$. Since I is conformal, its local effect must be an amplitwist. By considering the image of an arc of an origin-centred circle, deduce that $|(1/z)'| = 1/|z|^2$.
- 17 Consider the complex inversion mapping $I(z) = (1/z)$.
- (i) If $z = x + iy$ and $I = u + iv$, express u and v in terms of x and y .
 - (ii) Show that the Cauchy–Riemann equations are satisfied everywhere except the origin, so that I is analytic except at this point.
 - (iii) Find the Jacobian matrix, and by expressing it in terms of polar coordinates, find the local geometric effect of I .
 - (iv) Use (4.8) to show that the amplitwist is $-(1/z^2)$, just as in ordinary calculus, and in accord with the previous exercise. Use this to confirm the result of part (iii).
- 18 Recall Ex. 19, p. 210, where you showed that a general Möbius transformation

$$M(z) = \frac{az + b}{cz + d},$$

maps concentric circles to concentric circles if and only if the original family (call it $\mathcal{F}$) is centred at $q = -(d/c)$. Let $\rho = |z - q|$ be the distance from q to z , so that the members of $\mathcal{F}$ are $\rho = \text{const.}$

- (i) By considering orthogonal connecting vectors from one member of $\mathcal{F}$ to an infinitesimally larger member of $\mathcal{F}$, deduce that the amplification of M is constant on each circle of $\mathcal{F}$. Deduce that $|M'|$ must be a function of ρ alone.
- (ii) By considering the image of an infinitesimal shape that starts far from q and then travels to a point very close to q , deduce that at some point in the journey the image and preimage are *congruent*.
- (iii) Combine the above results to deduce that there is a special member I_M of $\mathcal{F}$ such that infinitesimal shapes on I_M are mapped to congruent image

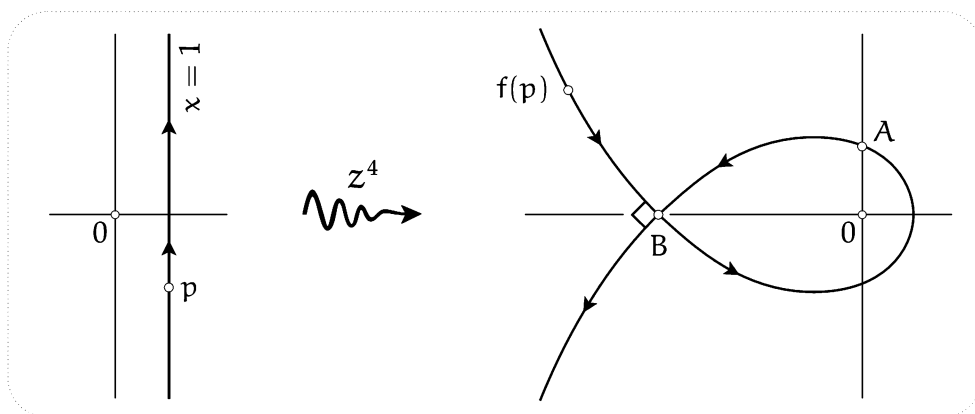
shapes on the image circle $M(I_M)$. Recall that I_M is called the *isometric circle* of M .

- (iv) Use the previous part to explain why $M(I_M)$ has the same radius as I_M .
- (v) Explain why $I_{M^{-1}} = M(I_M)$.
- (vi) Suppose that M is normalized. Using the idea in Ex. 16, show that the amplification of M is

$$|M'(z)| = \frac{1}{|c|^2 \rho^2}.$$

- 19 Consider the mapping $f(z) = z^4$, illustrated below. On the left is a particle p travelling upwards along a segment of the line $x = 1$, while on the right is the image path traced by $f(p)$.

- (i) Copy this diagram, and by considering the length and angle of p as it continues its upward journey, sketch the continuation of the image path.
- (ii) Show that $A = i \sec^4(\pi/8)$.
- (iii) Find and mark on your picture the two positions (call them b_1 and b_2) of p that map to the self-intersection point B of the image path.
- (iv) Assuming the result $f'(z) = 4z^3$, find the *twist* at b_1 and also at b_2 .
- (v) Using the previous part, show that (as indicated at B) the image path cuts itself at *right angles*.



- 20 The figure below is a copy of [2.9], p. 69.

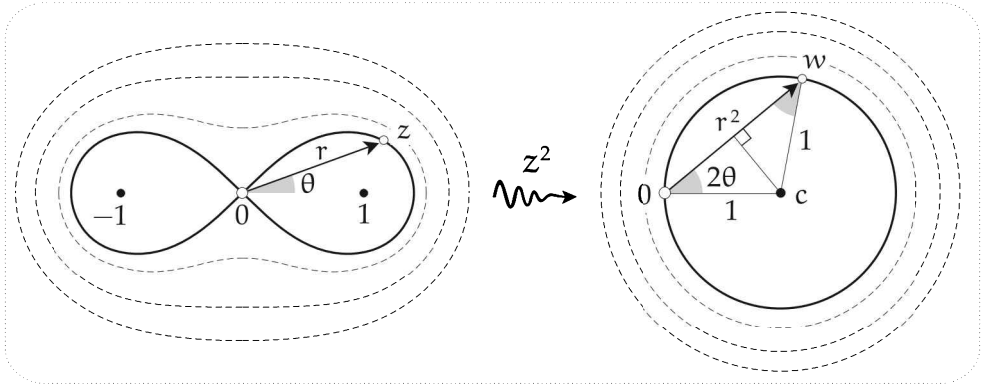
- (i) Show geometrically that if z moves distance ds along the lemniscate, increasing θ by $d\theta$, then $w = z^2$ moves a distance $4 d\theta$ along the circle.
- (ii) Using the fact that $(z^2)' = 2z$, deduce geometrically that $ds = 2 d\theta/r$.
- (iii) Using the fact that $r^2 = 2 \cos 2\theta$, show by calculation that

$$r dr = 2 \sqrt{1 - (r^4/4)} d\theta.$$

- (iv) Let s represent the length of the segment of the lemniscate connecting the origin to the point z . Deduce from the previous two parts that

$$s = \int_0^r \frac{dr}{\sqrt{1 - (r^4/4)}},$$

hence the name, *lemniscatic integral*.



- 21 (i) **Three-Dimensional SVD:** By extending the argument given in the text, show that in *three*-dimensional space the effect of a linear transformation is to stretch space in three mutually perpendicular directions (generally by three different factors), then to rotate it.
- (ii) Deduce that a mapping of three-dimensional space to itself is locally a three-dimensional amplitwist if and only if it maps infinitesimal spheres to infinitesimal spheres.
- (iii) Deduce that inversion in a sphere preserves the magnitude of the angle contained by two intersecting curves in space.
- (iv) Deduce that stereographic projection is conformal.

Remark: In stark contrast to the bountiful conformal mappings of the plane, Liouville and Maxwell independently discovered that the *only* angle-preserving transformation of space is an inversion, or perhaps the composition of several inversions.

